Voltage dividers & sensors

ENGINEERING WITH MR KELLY

Higher · Electronics & control · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ calculate the voltages in a voltage divider
- ☐ rearrange the voltage divider formula to find a resistor value
- ☐ describe how a sensing circuit works using LDR or thermistor resistance and Vout
- ☐ design a voltage divider to switch at a given light level or temperature

I DO — Worked example

Read how a full-mark answer is built

A frost alarm has a thermistor at the bottom of a voltage divider. Its resistance rises as the temperature falls.

1 Describe how the circuit switches the alarm on when it gets cold. (3 marks)

Full-mark answer each ✓ is one mark

As the temperature falls, the thermistor's resistance increases ✓. This means the voltage across the thermistor, Vout, increases ✓. When Vout reaches about 0.7 V the transistor switches on and current flows through the alarm ✓.

Why marks are not awarded

"When it gets cold the thermistor detects it and turns on the alarm."

0 marks: no reference to resistance or Vout.

WE DO — Guided practice

Use the starters and word bank to help you

A voltage divider has R1 = 10 kΩ and R2 = 15 kΩ across a 6 V supply.

2 Calculate V2. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Vs = 9 V, R1 = 1 kΩ, R2 = 2 kΩ. V2 is…
☐ 3 V ☐ 4.5 V ☐ 6 V ☐ 9 V

b) An LDR (2 kΩ) is at the bottom of a 9 V divider; the transistor must switch at 0.7 V. The top resistor is about…
☐ 2 kΩ ☐ 24 kΩ ☐ 12 kΩ ☐ 0.16 kΩ

c) In a divider, the larger resistor has…
☐ The smaller share of voltage ☐ The larger share of voltage ☐ No voltage ☐ The same voltage

d) A good description of a sensing circuit starts with…
☐ The relay ☐ The sensor's resistance changing ☐ The LED ☐ The battery

4 Key terms: write the letter of the matching description next to each term.

1. **Voltage divider rule**	____	**A** V1/V2 = R1/R2
2. **V out of a divider**	____	**B** Sensor resistance → Vin → transistor → relay → output.
3. **Sensor at the bottom of a divider**	____	**C** RA/RB = R1/R2; no voltage difference across the middle.
4. **LDR at the bottom**	____	**D** Vout rises as it gets darker (resistance rises).
5. **Describing a sensing circuit**	____	**E** Vout rises when the sensor's resistance rises.
6. **Balanced bridge**	____	**F** V2 = Vs × R2 ÷ (R1 + R2)

A thermistor is R2 in a voltage divider with a fixed resistor R1 = 4.7 kΩ and a 5 V supply. At 20 °C the thermistor is 2.2 kΩ.

5 Calculate Vout across the thermistor at 20 °C. (2 marks)

A light-sensing circuit needs V2 = 1.5 V from a 6 V supply. R2 (an LDR) is 1.2 kΩ at the switching light level.

6 Calculate the value of R1. (3 marks)

A Wheatstone bridge has two voltage dividers across 12 V: R1 = 10 kΩ, R2 = 10 kΩ; R3 = 8 kΩ, R4 = 12 kΩ.

7 Calculate the voltage between the two midpoints. (3 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?